



Superstore Sales Analysis Report

1. Project Overview

The objective of this project was to perform **Exploratory Data Analysis (EDA)** on the Superstore dataset to uncover business insights related to sales, profitability, customer behaviour, regional performance, product performance, and shipping efficiency. The project involved data cleaning, feature engineering, visualization, and business-oriented analysis to support data-driven decision-making.

2. Dataset Overview

The dataset contains transactional records from a retail superstore, including information about:

- Orders
- Customers
- Products
- Categories
- Regions
- Sales
- Profit
- Discounts
- Shipping

Several new features were engineered to improve analysis, including:

- Order Day, Month, Year
 - Ship Day, Month, Year
 - Shipping Time
 - Discount Category
 - Order Size
 - Month Names
-

3. Data Cleaning

The following preprocessing steps were completed:

- Checked and handled missing values
- Removed duplicate records
- Converted date columns into datetime format

- Verified data types
- Generated descriptive statistics
- Identified outliers
- Created additional business features

The dataset was successfully cleaned and prepared for analysis.

4. Exploratory Data Analysis

EDA was performed to answer several business questions using statistical summaries and visualizations.

The analysis included:

- Sales distribution
 - Profit distribution
 - Category-wise performance
 - Sub-category analysis
 - Regional performance
 - Customer purchasing behaviour
 - Shipping analysis
 - Discount impact
 - Monthly and yearly sales trends
 - Correlation analysis using heatmaps
-

5. Key Insights



Overall Business Performance

- The company generated over **\$2.3 million** in total sales.
 - Overall profit remained positive, indicating healthy business performance.
 - Average order value shows customers make moderate-sized purchases.
-



Technology is the Best Performing Category

Technology generated the highest sales and the highest profit among all categories.

This indicates that technology products contribute significantly to overall business growth.

Furniture Needs Improvement

Although Furniture contributes substantial sales, its profit margins are considerably lower due to higher discounts and operational costs.

Regional Performance

The **West** region consistently achieved the highest sales and profits.

The **South** and **Central** regions generated comparatively lower profits, suggesting opportunities for business improvement.

Sales Increased Every Year

Year-over-year analysis showed continuous growth in sales and profit.

This indicates increasing customer demand and successful business expansion.

Discounts Reduce Profitability

The analysis revealed that:

- Higher discounts generally reduce profit.
- Products with heavy discounts often generate losses.
- Moderate discounts produce better financial performance.

This suggests that discount strategies should be optimized instead of offering large discounts indiscriminately.

Shipping Performance

By creating the **Shipping Time** feature, shipping efficiency was analyzed.

The analysis helps identify:

- Average delivery time
 - Shipping delays
 - Most frequently used shipping modes
 - Opportunities to improve customer satisfaction
-

Customer Analysis

The project identified:

- Top customers by sales

- High-value customers
- Repeat customers
- Customer purchasing behaviour

These insights can support customer retention and loyalty programs.

Product Analysis

The analysis highlights:

- Best-selling products
- Highest-profit products
- Low-performing products
- Loss-making products

These findings can assist inventory optimization and pricing decisions.

Correlation Analysis

A correlation heatmap was used to examine relationships among numerical features.

The analysis showed:

- Sales and Profit have a positive relationship.
 - Discounts negatively impact Profit.
 - Shipping Time has minimal correlation with Sales.
-

6. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Increase investment in Technology products since they generate the highest revenue and profit.
 - Reduce excessive discounts to improve profitability.
 - Improve pricing strategies for Furniture products.
 - Replicate successful sales strategies from the West region in lower-performing regions.
 - Develop customer loyalty programs targeting repeat and high-value customers.
 - Optimize inventory by focusing on profitable products.
 - Improve shipping efficiency to enhance customer experience.
-

7. Skills Demonstrated

This project demonstrates practical skills in:

- Data Cleaning
- Data Preprocessing
- Feature Engineering
- Exploratory Data Analysis (EDA)
- Data Visualization
- Business Intelligence
- Customer Analytics
- Sales Analytics
- Profitability Analysis
- Python Programming
- Pandas
- NumPy
- Matplotlib
- Seaborn

8. Conclusion

This project successfully transformed raw retail transaction data into meaningful business insights through systematic data cleaning, feature engineering, and exploratory analysis. The findings reveal key trends in sales, profitability, customer behaviour, and regional performance that can support strategic business decisions. The project demonstrates how data analytics can be leveraged to optimize pricing, improve customer satisfaction, increase operational efficiency, and drive business growth.

Overall, this analysis provides a strong foundation for future predictive modeling, sales forecasting, customer segmentation, and business dashboard development using tools such as Power BI or Tableau.